



**NATIONAL
YANG MING CHIAO TUNG
UNIVERSITY**

Homework Report #1

HW Report Submitted By:

Name: _____

RollNo: _____

Submitted To:

Course title: Programming Language

Course number: BEBE10011

Department: Biomedical Engineering

HW Date:

Submission Date:

Marks & Signature

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Title:

Write a program to take input of name, rollno and marks obtained by a student in 5 subjects each have its 100 full marks and display the name, rollno with percentage score secured.

Problem Analysis:

Based on the problem, it is required to get the input of name, roll number and marks in 5 subjects of a student. The program should display the name; roll number and percentage of marks secured by that student as output. The input variables shall be: name, rollno, msub1, msub2, msub3, msub4, msub5. We need to calculate percentage of marks obtained. So the variable 'score' holds the percentage to be displayed.

Percentage of marks obtained = $\frac{\text{total marks on 5 subjects}}{\text{total full marks}} \times 100$

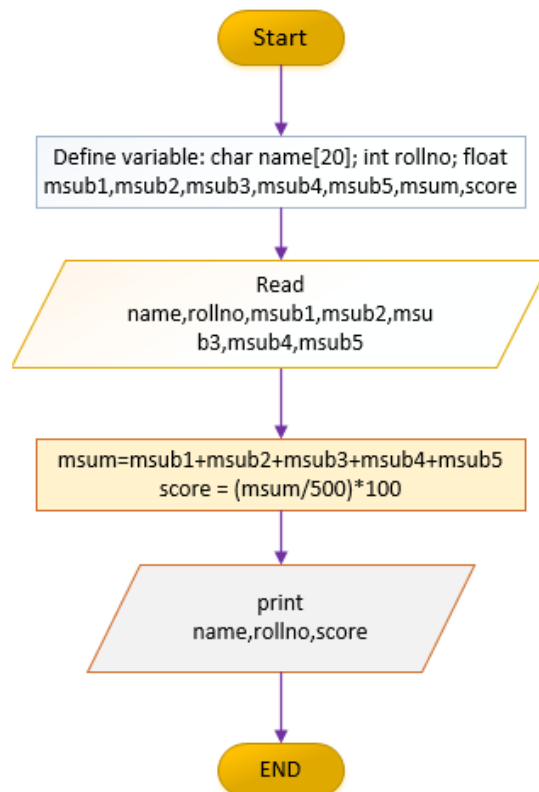
Hence, $\text{msum} = \text{msub1} + \text{msub2} + \text{msub3} + \text{msub4} + \text{msub5}$; $\text{Score} = \frac{\text{msum}}{500} \times 100$

Input variables	Processing variables/calculations	Output variables	Necessary header files/functions/macros
Name (char type) rollno (int) msub1, msub2, msub3, msub4, msub5 (float)	msum (float)	name (char type) rollno (int) score(float)	stdio.h conio.h scanf() &printf() for formatted i/o.

Algorithm:

1. Start
2. Define variables: name, rollno, msub1, msub2, msub3, msub4, msub5, msum, score
3. Take input from keyboard for all the input variables
4. Calculate the sum of marks of 5 subjects and also calculate the percentage score as:
$$\text{Msum} = \text{msub1} + \text{msub2} + \text{msub3} + \text{msub4} + \text{msub5}; \text{Score} = \frac{\text{msum}}{500} \times 100$$
5. Display the name, roll number and percentage score.
6. Stop

Flowchart:



Code:

```
/******  
  
Online C Compiler.  
Code, Compile, Run and Debug C program online.  
Write your code in this editor and press "Run" button to compile and execute it.  
  
*****/  
#include<stdio.h>  
#include<conio.h>  
int main(void)  
{  
    char name[20];  
    int rollno;  
    float msub1, msub2, msub3, msub4, msub5, msum, score;  
    printf("Enter Name of Student: ");  
    scanf("%[^\n]", name); /*can use scanf("%s",name) but it reads single word only.*/  
    printf ("\nRoll Number: ");  
    scanf("%d", &rollno);  
    printf ("\nEnter Marks in 5 Subjects:\n");  
    scanf("%f%f%f%f%f", &msub1, &msub2, &msub3, &msub4, &msub5);  
    msum = msub1 + msub2 + msub3 + msub4 + msub5;  
    score = msum/500*100;  
    printf("\nName of Student: %s", name);  
    printf("\nRoll Number: %d", rollno);  
    printf ("\nPercentage Score Secured: %2.2f%c", score, '%');  
}
```

Output (Compilation, Debugging & Testing):

```
Enter Name of Student: Ken Chen  
Roll Number: 522  
Enter Marks in 5 Subjects:  
45.5  
50  
63  
76  
62.5  
Name of Student: Ken Chen  
Roll Number: 522  
Percentage Score Secured: 59.40%
```

Discussion & Conclusion:

In this first lab of C/C++ Programming, based on the focused objective(s) to understand about C data types with formatted input/output functions, the additional lab exercises made me more confident towards the fulfillment of the objectives.